

Bihan Banerjee

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EXPERIENCE

MLOps Intern

June 2025 – July 2025

DCG AILABS Academy Pvt Ltd

Kolkata, India

- Engineered a real-time sentiment analysis system on the IMDb dataset with **50K+** reviews using Flask.
- Developed and deployed an AI model with **TF-IDF vectorization** and containerized with Docker, achieving 89% accuracy.
- Automated streaming workflows using **Kafka and Zookeeper** for live inference and dynamic feedback dashboards.

General Secretary, Mozilla Firefox Club VIT

July 2024 – Present

Vellore Institute of Technology

Vellore, India

- Led planning and execution of multiple technical and non-technical events and hackathons for **300+** students.
- Coordinated with cross-functional teams for sponsorships, logistics, and technical workshops.
- Anchored **50+ events** and sessions for juniors, guiding them through technical workshops and club activities.

PROJECTS

CyberX – AI-Powered Cybersecurity Simulator | Python, MERN, Reinforcement Learning

2025 – 2026

- Architected a modular **red-team/blue-team cybersecurity framework** integrating a custom pentesting tool suite with cryptographic, forensic and defensive monitoring modules with a total of **25+ tools**.
- Built real-time attack intelligence dashboards featuring network details and **geospatial globe-based threat visualization** capable of rendering **1,000+** live events with **<2s refresh intervals**.
- Developed system-level monitoring components tracking CPU, memory, and network metrics with 1s sampling intervals while **maintaining <5% monitoring overhead**.
- Simulated attacker-defender interactions over **100-step episodes** using structured **honeypot telemetry**.
- Implemented **PPO-based** attacker and defender agents trained via adversarial self-play across **500K+ simulation timesteps**, improving attack detection rates over baseline policies.

Secure Code – Vulnerability Scanner | React, Flask, TypeScript, MongoDB

2025

- Designed and developed an AI-driven secure code analysis platform to detect and **auto-remediate vulnerabilities** in Python and JavaScript code generated by LLMs using **CodeT5, GPT-Neo, and CodeGPT-small**.
- Integrated static analysis tools (**Bandit, Semgrep, SonarQube**) to perform multi-layered vulnerability detection mapped to **100+ CWE vulnerability patterns**.
- Engineered an AI-based secure code transformation pipeline generating patched code with **diff visualization**, reducing insecure patterns by **70–85% in benchmark evaluations**.
- Built a scalable full-stack React (TypeScript) + Flask system supporting multi-file uploads, modular scanning, real-time vulnerability dashboards, and secure **bcrypt-based MongoDB authentication**.

Serenity Gardens – Family Garden Webapp | MERN, TypeScript, Razorpay

2025

- Architected and developed a full-stack e-commerce platform with public pages (landing, gallery, reviews) and a secure shopping portal using React, Node.js, Express, and MongoDB.
- Implemented role-based authentication using **JWT and bcrypt**, enabling admin-controlled catalog management and user-specific cart and order workflows.
- Engineered dynamic product catalog, cart persistence, and checkout system supporting Cash on Delivery and **Razorpay payment integration**, with automated order recording in MongoDB.
- Built **responsive UI** components and **RESTful API** integrations, ensuring seamless state management, secure transactions, and scalable backend operations.

NeuroPharm – AI-Driven Drug Discovery Platform | React, Node.js, Python

2024

- Built an AI-driven drug discovery platform integrating **AlphaFold and ChEMBL APIs** to retrieve and process **1,000+ protein and ligand records** for automated screening.
- Engineered Node.js + Python backend for ligand ranking, structured dataset management, and protein-ligand interaction analysis with **interactive 3D visualization**.
- Developed a React-based frontend enabling real-time exploration of molecular datasets and dynamic structural rendering.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++, SQL

Frameworks: React, Node.js, Flask, Express, Vite, TailwindCSS

Cybersecurity Tools: Burp Suite, Kali Linux, Wireshark, Metasploitable, Semgrep, Bandit, SonarCloud

Developer Tools: Git, MongoDB, Kafka, Docker, VS Code, Postman

EDUCATION

Garden High School

Kolkata, India

Council for the Indian School Certificate Examinations (CISCE)

2008 – 2023

Vellore Institute of Technology (VIT)

Vellore, India

B.Tech in Computer Science and Engineering (Information Security)

2023 – Present (Expected 2027)